

ISINDU WIJESINGHE

ASSOCIATE SOFTWARE ENGINEER | FULL-STACK • MOBILE • CLOUD • APPLIED AI/ML

Colombo, Sri Lanka | LinkedIn | GitHub | Portfolio

01 PROFILE

Associate Software Engineer and BSc (Hons) Information Technology undergraduate at SLIIT. Hands-on experience spans mobile and web application delivery, REST APIs, database-backed systems, real-time features, geospatial applications, software release support, and full-stack product development. Current technical work also includes cloud architecture and applied AI/ML research, with an emphasis on practical systems, measurable trade-offs, and deployable software.

02 PROFESSIONAL EXPERIENCE

Associate Software Engineer — LAUGFS Holdings

2026 - Present

Colombo, Sri Lanka

- Develop and maintain business-facing web and mobile application features across Flutter/Dart, modern JavaScript front ends, and API-driven back-end workflows within existing enterprise codebases.
- Work across REST integration, database-backed application flows, defect diagnosis, Android build/configuration issues, network/API troubleshooting, and regression verification during development and QA cycles.
- Prepare structured QA and UAT mobile releases, including versioned APK handovers, release notes, installation/testing guidance, and supporting technical communication for test and project teams.
- Contribute to implementation, debugging, technical documentation, release coordination, and post-fix verification while collaborating with QA and project stakeholders.

03 SELECTED ENGINEERING PROJECTS

FileGrip — Online File Processing Platform

2026 - Present

Production web application | filegrip.cc/

PDF • DOCUMENT • IMAGE PROCESSING • PRIVACY-AWARE FILE WORKFLOWS

- Developed a production file-utility platform for PDF, document, and image workflows, covering merge, split, compression, rotation, page deletion/reordering, protection/unlocking, redaction, signing, watermarking, and common conversion operations.
- Designed a low-friction upload-process-download experience with no-signup basic tools, temporary processing, automatic file deletion/expiry, and a clear privacy-first handling model.
- Organized capabilities into PDF, conversion, image, security, and editing tool groups so individual utilities remain focused while sharing a consistent user flow and product experience.

UniLocate — Smart Geo-Fenced Campus Intelligence System

Feb 2026 - Apr 2026

Team project | Student application, admin dashboard, APIs and infrastructure

REACT NATIVE / EXPO • REACT / MUI • NODE.JS / EXPRESS • POSTGRESQL / POSTGIS • REDIS • DOCKER

- Led major student-app and geo-intelligence implementation for a campus platform combining real-time occupancy, location-aware navigation, anonymous complaints, lost & found, student services, and role-based administration.
- Implemented geofenced location handling, zone matching, on-device location history, map/search interactions, campus-boundary logic, occupancy refresh flows, and PostGIS-backed location-event processing.
- Integrated API and infrastructure components using Express, PostgreSQL/PostGIS, Redis, and Docker, while supporting modular services and mobile-to-backend communication.
- Explored voice-assisted interaction for accessibility, including location and lost-item command scenarios intended to reduce friction for visually impaired users.

LoopCart — Full-Stack Retail & POS Management System

Oct 2025 - Nov 2025

Personal full-stack engineering project

ASP.NET CORE • REST API • REACT • REDUX • JAVASCRIPT • SQL

- Built a complete retail/POS workflow covering checkout, product and category management, inventory control, order history, discounts/promotions, user roles, search, filtering, pagination, and administrative analytics.
- Designed and consumed REST endpoints between the ASP.NET Core back end and React front end, with Redux-based application state used to coordinate shared product, order, and user data.
- Used the project to practice layered application architecture, reusable UI composition, validation, API error handling, and mapping day-to-day retail processes into maintainable software features.

03 SELECTED ENGINEERING PROJECTS — CONTINUED

MotionChat — Real-Time Customer Support Platform

Jul 2025

Full-stack real-time web application

REACT • TAILWIND CSS • NODE.JS • EXPRESS • MONGODB • SOCKET.IO • JWT • EMAIL OTP

- Developed guest and registered-user support chat flows with live agent assignment, session management, topic routing, message editing/deletion, post-chat ratings, and agent/admin controls.
- Implemented two-way real-time communication with Socket.IO, JWT-based authentication, email OTP flows, and MongoDB-backed user/session data.
- Built an agent dashboard for ongoing/responded chats, approvals, customer ratings, and session visibility while maintaining a responsive animated React interface.

ShopCart — Smart Grocery Logistics Management System

Apr 2025 - Jul 2025

MERN full-stack project

MONGODB • EXPRESS.JS • REACT • NODE.JS • JWT • REST APIs

- Implemented customer and administrative workflows for authentication, catalog browsing/search, cart and checkout, order history, inventory management, stock control, and order-status progression.
- Created REST API interactions and role-separated management flows for products, orders, delivery status, and basic operational sales/stock visibility.

04 APPLIED AI/ML & RESEARCH

GeoNex — AI-Supported Landslide Early Warning Research

Ongoing 2026

Final-year research | sensing, edge intelligence and risk analytics

PYTHON • EDGE / IoT • ESP32 • RASPBERRY PI • MQTT / LoRa • TFLITE / TINYML CONCEPTS • TIME-SERIES ML

- Researching a low-cost early-warning architecture that combines distributed environmental sensing, edge-side processing, centralized monitoring, and explainable risk outputs for landslide-prone environments.
- Exploring hybrid rule-based and machine-learning approaches for sensor validation, anomaly/fault identification, credibility scoring, and risk classification under constrained edge-compute conditions.
- Evaluating model families such as Random Forest, XGBoost, SVM, and recurrent approaches for risk/anomaly modelling, with attention to memory, latency, reliability, and deployment constraints.

Efficient Language-Model Architecture Study

Ongoing 2026

Independent AI architecture study

ATTENTION / KV-CACHE EFFICIENCY • REPRESENTATION LEARNING • COMPUTE / MEMORY ANALYSIS • COMPACT-MODEL EXPERIMENTATION

- Studying architectural choices used in modern language models and how attention, cache design, representation quality, and dimensionality affect inference compute and memory cost.
- Working toward small-scale experimental implementations that preserve the structure of larger architectures while remaining practical for limited local compute.

05 EDUCATION

BSc (Hons) in Information Technology — Specializing in Information Technology

2023 - 2027 (Expected)

Sri Lanka Institute of Information Technology (SLIIT)

- Applied coursework and project work across software engineering, mobile and web development, cloud computing, databases, data warehousing/business intelligence, distributed systems, and AI/ML-related research.

06 CERTIFICATIONS

Amazon Web Services (AWS) — AWS SimuLearn: Databases in Practice	Sep 2026
Amazon Web Services (AWS) — AWS SimuLearn: First NoSQL Database	Sep 2026
Amazon Web Services (AWS) — AWS SimuLearn: Computing Solutions	Aug 2026
Amazon Web Services (AWS) — AWS SimuLearn: File Systems in the Cloud	Aug 2026
Amazon Web Services (AWS) — AWS SimuLearn: Highly Available Web Applications	Aug 2026
Amazon Web Services (AWS) — AWS SimuLearn: Networking Concepts	Aug 2026
HackerRank — Frontend Developer (React)	Nov 2025
HackerRank — SQL (Basic)	Nov 2025
Postman — API Fundamentals Student Expert	Sep 2025
HackerRank — Software Engineer (Intern)	Nov 2023

07 TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C#, PHP, Dart, SQL, HTML, CSS

Frontend & Mobile: React, React Native, Expo, Redux, Vue.js, Flutter, Vite, Tailwind CSS, Material UI, Framer Motion

Backend & APIs: Node.js, Express.js, NestJS, ASP.NET Core, Laravel, REST APIs, Socket.IO, JWT, RBAC, API integration

Data & Platform: PostgreSQL, PostGIS, MongoDB, SQL Server, Redis, Docker, AWS fundamentals, Git/GitHub, Postman, Vercel

Data / BI & Tools: SSIS, SSAS, Power BI, Android Studio, Visual Studio, VS Code, QGIS